

AFRE 991: Advanced Data Analytics and Frontier Methods for Applied Economists

**(3 credits: Tue Thu: 10:20-11:40 AM – Berkey Hall 214)
Department of Agricultural, Food, and Resource Economics**

Course Description:

Advanced topics such as price analysis, finance, risk and modeling techniques, agri-food systems, environmental economics and management, and agricultural and natural resource development and policy.

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Office Hours: Thursdays 11:40AM-1:10PM and upon request (in person and [Zoom](#))
Course Website: <https://github.com/afre-msu/AFRE-891-991-FS26>

Prerequisites:

AFRE 802 and AFRE 835 or equivalents. In order to focus energy on data science skills and forefront methods, this course assumes graduate-level knowledge of econometric methods and statistics.

Course Overview:

Training in data analytics for applied economists and social scientists with emphasis on data acquisition, advanced manipulation and visualization techniques, and spatial data methods. Coverage of frontier empirical methods including machine learning and synthetic control methods.

This course is designed as a primer on modern data science and applied economic analytical techniques for Ph.D. students. At a high level, this course aims to provide a core competency in practical data analytics skills at all stages of the research supply chain: from data acquisition to data cleaning, manipulation, and visualization, as well as forefront synthetic control methods and techniques like machine learning, spatial analysis, and big data analysis that lie outside the scope of other quantitative methods courses. Many of these skills are essential components of the applied economists' research workflow but are nevertheless often left out of the core graduate program curriculum. My intention is to provide training in the empirical tools that will help students make the transition from consumers of graduate coursework to producers of high-quality, modern, empirical economic research.

In short, this is the class I wish I could have taken before starting my independent graduate research.

Instructional Objectives:

Upon successful completion of this course, students should be able to

- Utilize modern statistical programming methods for data cleaning, merging, iteration, and manipulation of complex data structures (including geospatial data)
- Effectively structure research project workflows and generate reproducible code
- Conduct version control effectively using Github
- Create professional-quality data visualizations
- Use web scraping techniques to compile datasets from websites with wide ranges of webpage structures

- Estimate fixed effects and instrumental variable models quickly in R, compile high-quality results tables and generate figures from estimation results
- Effectively utilize synthetic control and machine learning methods, understand how these empirical methods differ from traditional econometric models, and when these techniques are appropriate

You will meet the objectives listed above through the combination of the following activities in this course:

- Lectures with students actively replicating code and analyses
- Homework assignments in which students will conduct original analysis or replicate and extend existing research papers
- Development and completion of an original research project

Delivery Format

Lectures will be taught in-person on Tuesdays and Thursdays from 10:20 – 11:40am in Berkey Hall 214. I will also record all lectures and upload recordings to the class D2L for later viewing (for those who are unable to make class on a given day or for later review) within 48 hours of class time.

Class Webpage (Github)

I will exclusively be using our course-specific repository on [Github](https://github.com/afre-msu/AFRE-891-991-FS26) (<https://github.com/afre-msu/AFRE-891-991-FS26>) to disseminate all course materials. Early in the course I will provide instruction as to how to register for Github and use Github Desktop to automatically sync all course materials to your computer as they are published/updated.

Required Course Materials

This course has no required textbooks; all necessary materials will be provided through the weekly lecture slides available on the course website.

Required Software

Please plan to have all required software (R, RStudio, and Windows Rtools/macOS R toolchain) noted below installed before our first lecture as we will jump right into things. If you run into installation issues, please reach out to me as soon as possible. I will also be available in office hours during the first week of the semester for additional troubleshooting.

1. R

The class will be taught primarily using the statistical programming language R. While R can be used through the console, I will be teaching with the RStudio integrated development environment (IDE), and highly recommend you install it as well.

Why R? Beyond the immediate benefits of being free and open-source, R has been designed from the ground-up for data science and accordingly is heavily utilized in industry, increasingly used for academic research, and has a rich user community. When it comes to emerging methods, many tools appear in R before other platforms. R also has tremendous flexibility in the types of data it can read and the analytical methods it can employ, allowing for simplified workflows. As well, R can be [much faster](#) than Stata in typical econometrical use cases.

You can download R for free from the [Comprehensive R Archive Network \(CRAN\)](#) for Linux, macOS, and Windows.

RStudio Desktop (also free) is available [here](#) for a wide range of operating systems.

2. Tools for R Packages

Throughout the course we will be making use of R packages, bundles of functions and documentation that build on base R and allow R to do... a lot of stuff. While many are available centrally through CRAN and can be installed directly, some packages that we will use in this class are only available through Git repositories and require additional tools for installation. Please download and install the following tools for your OS:

- Windows: install [Rtools](#).
- macOS: install the [macOS R toolchain](#).
- Linux: no additional software needed
- Jailbroken TI-83 calculator: I have some bad news...

A note: the main goal of this class, data analytical literacy, is not specific to R. While we're developing our skills this semester through R, the ability to conceptualize analytical workflows and solve data challenges is core to data analysis in Stata, Python, ArcGIS, and any other platform you may use in the future or may deem optimal for a given task.

I will indicate in-class and on lecture slides when any additional software needs to be installed.

Optional Textbooks and References

There is no required textbook for this class. The lecture notes are designed to be as self-contained as possible and draw from various resources (huge shoutouts to Nick Hagerty at Montana State, and Ed Rubin and Grant McDermott at U Oregon). A few of my inspiration sources are provided below as well as other recommendations for expanded reading – all are available for free online.

- [R for Data Science](#) by Hadley Wickham and Garrett Grolemund
- [Advanced R](#) by Hadley Wickham
- [Data Visualization: A Practical Introduction](#) by Kieran Healy
- [Geocomputation with R](#) by Robin Lovelace, Jakub Nowosad, and Jannes Muenchow
- [Spatial Data Science with Application in R](#) by Edzer Pebesma and Roger Bivand
- [An Introduction to Statistical Learning](#) by Gareth James, Daniela Witten, Trevor Hastie, and Robert Tibshirani

As the above shows, there is a wealth of incredible resources available online and I encourage you to make use of any additional resources that you want. If you find a specific resource of particular benefit, I would love to hear about it to hopefully integrate more of its perspective in the future.

Assessments and Grading

Assessments for the course fall into one of two types:

1. Homework Assignments

Students will complete 6 homework assignments over the course of the semester. Homework assignments are designed to provide hands-on practice with the analytical skills covered every 2-3 weeks. Many will take the form of replicating and extending existing studies, manipulating and analyzing a provided dataset, or

compiling an original dataset. I encourage you to work with your fellow students to complete these assignments, but every student is expected to submit their own unique version of the assignment; the nature and extent of any collaboration with other students should be thoroughly documented at the start of a submission.

Most homework assignments require a combination of code, output from code, and written answers. I will grade answers for completeness and accuracy. Code will be further evaluated for efficiency: considerable redundancy and excessive unnecessary steps will be marked down, even if it “runs.”

All homework assignments will be distributed through GitHub using [Classroom 50](#) (replacement for Github Classrooms, RIP). Early in the course you should receive an email invitation with instructions on how to join the course classroom.

Unexcused late assignments will be given a grade of zero. If you are worried that you will be unable to submit an assignment on time, make sure to communicate that to me as far in advance as possible.

2. Research Project and Presentation

Throughout the semester, students will work on an independent research project. Midway through the semester, students will submit a brief prospectus outlining the proposed project and how it relates to both the course content and your graduate research portfolio. This can take the form of developing a novel dataset, replicating a paper in your field and extending it with the addition of new data or empirical techniques, or using covered course methods to tackle a desired research question (i.e. forming analyses for a research paper). Consider this an opportunity to incubate one of your ideas for a potential third-year paper, or make progress on a new research question.

At the end of the semester, students will give a 10-minute, conference-style presentation on your research project. Each presenter will be assigned a discussant, who will review the final project submission and provide a brief 3-minute oral feedback following the project’s presentation.

In addition to the final research presentation, students will submit **replication packages** (code and utilized data) through GitHub by end of day November 30 that will allow for the replication of all figures and tables presented in the research presentation by myself and your discussant. If there are privacy or data sharing concerns regarding your data, students should make arrangements with me in advance of the final presentation.

Note: I am not expecting this to be a completed research project. Rather, consider this an opportunity to trial an idea that you have and see if you are able to acquire the data or perform a preliminary version of the desired analyses.

Course Grade Determination

Final course grades will be calculated as follows:

Homework Assignments	6 at 13.33% each = 80%
Final Project: Proposal	3%
Final Project: Replication Package	7%
Final Project: Presentation	7%
Final Project: Discussant	3%

Percentage grades will be converted to the university 4-point grading scale based on the following:

4.0 = 92.0% to 100%	2.0 = 70.0% to 74.9%
3.5 = 85.0% to 91.9%	1.5 = 65.0% to 69.9%
3.0 = 80.0% to 84.9%	1.0 = 60.0% to 64.9%
2.5 = 75.0% to 79.9%	0 = Less than 60.0%

Office Hours:

I will hold office hours from 11:40am-1:10pm on Thursdays to assist with questions. I will be available both in person immediately after class on the walk back to Ag Hall and over [Zoom](https://msu.zoom.us/j/95806445345) (<https://msu.zoom.us/j/95806445345>, pass: gogreen) shortly thereafter. If that time does not work for you, please send me an email and we can schedule a different time to meet.

Attendance, Participation, and Engagement:

During all classes, I expect students to be fully engaged and adequately prepared to follow along with coding and covered methods. Bringing an appropriately set up laptop with you is essentially required to keep up with course materials. I encourage students to ask questions of the instructor and their peers.

Note that, while I will not take attendance, the course moves very quickly and builds upon itself. Attendance and engagement is necessary for success in this class. If you are unable to attend class on a particular day, make sure to work through the lecture slides and watch the lecture recording prior to our next class so that you do not fall behind.

AI Use:

This course is designed to develop your coding muscles. Like most things worth doing, developing skill in coding and research workflows requires practice – both with the analytical steps themselves as well as adapting/problem solving when things break. Being able to bang your head against the proverbial wall, problem solve, and figure out solutions to challenges is a huge value-add and a clearcut way to differentiate yourself from other job candidates (and an absolutely essential skill if you want to survive in an academic career). Beyond this course, the way we get jobs after graduation is not just by producing quality research but by convincing a hiring committee or seminar audience that the decisions we made throughout were the best choices to make. How can you defend your decision-making process if AI made the choices for you?

A learning objective of this class is developing the expertise to properly evaluate genAI output. This requires fundamental knowledge of the topic or object considered (which I'd argue is the whole point of pursuing a graduate degree), which you develop through practice. As such, the use of genAI in this course is restricted in the following ways:

- Homework assignments: AI is **not permitted** for use on homework assignments. If you get stuck with a function, consult the R documentation, the package website, discuss with your fellow students, or search for a general version of the workflow/issue on [StackOverflow](#) and adapt the provided example code. Any problems submitted with hallucinated AI code that doesn't compile will receive scores of zero and receive a warning; any future violations will receive scores of zero on the entire assignment.
- Final project: AI is permitted for the following functions: drafting presentation slides, drafting code (when you have developed sufficient knowledge to evaluate output for the given task). All code submitted in the replication package must be commented by hand and sufficient to demonstrate that you know what each step in your workflow is accomplishing.

For more on navigating a PhD in the age of genAI, I highly recommend reading [this article by UM professor Eytan Adar](#).

OTHER COURSE POLICIES:

Diversity, Equity, and Inclusion: Diversity, Equity and Inclusion are important, interdependent components of everyday life in the College of Agriculture and Natural Resources (CANR) and are critical to our pursuit of academic excellence. Our aim is to foster a culture where every member of CANR feels valued, supported and inspired to achieve individual and common goals with an uncommon will. This includes providing opportunity and access for all people across differences of race, age, color, ethnicity, gender, sexual orientation, gender identity, gender expression, religion, national origin, migratory status, disability / abilities, political affiliation, veteran status and socioeconomic background.

Academic Honesty: Article 2.3.3 of the [Academic Freedom Report](#) states that "The student shares with the faculty the responsibility for maintaining the integrity of scholarship, grades, and professional standards." In addition, the (insert name of unit offering course) adheres to the policies on academic honesty as specified in General Student Regulations 1.0, Protection of Scholarship and Grades; the all-University Policy on Integrity of Scholarship and Grades; and Ordinance 17.00, Examinations. (See [Spartan Life: Student Handbook and Resource Guide](#) and/or the MSU Web site: www.msu.edu.) Therefore, unless authorized by your instructor, you are expected to complete all course assignments, including homework, lab work, quizzes, tests and exams, without assistance from any source. You are expected to develop original work for this course; therefore, you may not submit course work you completed for another course to satisfy the requirements for this course. Also, you are not authorized to use the www.allmsu.com Web site to complete any course work in this course. Students who violate MSU academic integrity rules may receive a penalty grade, including a failing grade on the assignment or in the course. Contact your instructor if you are unsure about the appropriateness of your course work. (See also the MSU [Academic Integrity](#) webpage.)

Limits to Confidentiality. Assignments, code, and other materials submitted for this class are generally considered confidential pursuant to the University's student record policies. However, students should be aware that University employees, including instructors, may not be able to maintain confidentiality when it conflicts with their responsibility to report certain issues to protect the health and safety of MSU community members and others. As the instructors, we must report the following information to other University offices (including the Department of Police and Public Safety) if you share it with one of us:

- Suspected child abuse/neglect, even if this maltreatment happened when you were a child,
- Allegations of sexual assault or sexual harassment when they involve MSU students, faculty, or staff, and
- Credible threats of harm to oneself or to others.

These disclosures may trigger contact from a campus official who will want to talk with you about the incident that you have shared. In almost all cases, it will be your decision whether you wish to speak with that individual. If you would like to talk about these events in a more confidential setting you are encouraged to make an appointment with the MSU Counseling Center.

Accommodations for Students with Disabilities (from the Resource Center for Persons with Disabilities (RCPD): Michigan State University is committed to providing equal opportunity for participation in all programs, services and activities. Requests for accommodations by persons with disabilities may be made by contacting the Resource Center for Persons with Disabilities at 517-884-RCPD or on the web at rcpd.msu.edu. Once your eligibility for an accommodation has been determined, you will be issued a Verified Individual Services Accommodation ("VISA") form. Please present this form to the instructor at the start of the term and/or two weeks prior to the accommodation date. Requests received after this date may not be honored.

Commercialized Lecture Notes: Commercialization of lecture notes and university-provided course materials is not permitted in this course without expressed permission by the lecturer.

Tentative Course Calendar

Week	Tuesday	Thursday	Topics	Final Project	Assessments
1	September 1 Class 1 Lecture	September 3 Class 2 Lecture	Course Introduction R Basics		
2	September 8 Class 3 Lecture	September 10 Class 4 Lecture	Version Control with GitHub R Markdown		
3	September 15 Class 5 Lecture	September 17 Class 6 Lecture	Data Wrangling Joining Data		Assignment 1 Due Friday, September 18
4	September 22 Class 7 Lecture	September 24 Class 8 Lecture	Data Tidying with <i>tidyr</i> Data Cleaning		
5	September 29 Class 9 Lecture	October 1 Class 10 Lecture	Data Cleaning, Continued		Assignment 2 Due Friday, October 2
6	October 6 Class 11 Lecture	October 8 Class 12 Lecture	Data Visualization		
7	October 13 Class 13 Lecture	October 15 Class 14 Lecture	Data Acquisition Scraping Static Websites Scraping Dynamic Websites	Proposal Due Friday, October 16	Assignment 3 Due Friday, October 16
8	October 20	October 22	Function Writing		

	Class 15 Lecture	Class 16 Lecture	Vectorization and Paralellization	
9	October 27 No Class Fall Break	October 29 Class 17 Lecture	Data Acquisition through APIs Intro to Programming Iteration	
10	November 3 Class 18 Lecture	November 5 Class 19 Lecture	Function Writing Vectorization and Paralellization Intro to Regression	Assignment 4 Due Friday, November 6
11	November 10 Class 20 Lecture	November 12 Class 21 Lecture	Fast Fixed Effects and IV Regression Tables and Figures from Regression Output Causal Inference Methods in R	
12	November 17 Class 22 Lecture	November 19 Class 23 Lecture	Synthetic Control Methods in R Introduction to Spatial Data Vector Data in R	Assignment 5 Due Friday, November 20
13	November 24 Class 24 Lecture	November 26 No Class Holiday	Raster Data in R Joining Raster and Vector Data Spatial Regression Methods	
14	December 1 Class 25 Lecture	December 3 Class 26 Holiday	Intro to Machine Learning and Classification Model Selection and Regularization Regression Trees and Forest-Based Methods	Material Submission Due Sunday, December 6
15	December 8	December 10	Machine Learning for	

	Class 27	Class 28	Causal Treatment Effects	Discussant Response due Friday, December 11	Assignment 6 Due Monday, December 14
	Lecture	Lecture			
16	Final Presentations - Timing TBD				